

Housing Price Prediction – Findings Summary

This project focused on analyzing a housing dataset and building machine learning models to predict house prices based on various property characteristics.

The dataset was first explored to understand its structure, identify the target variable (**price**), and check for missing values. Data cleaning involved removing duplicate records and converting categorical variables (such as road access, guest room availability, basement presence, and furnishing status) into numerical form using one-hot encoding. This preprocessing step ensured that the dataset could be used effectively by machine learning algorithms.

For model development, the cleaned dataset was divided into training and testing sets using an 80:20 split. Two regression models were trained and evaluated: **Linear Regression** and **Random Forest Regression**. Model performance was measured using standard regression metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

The analysis revealed that **property area** and **number of bathrooms** were among the most influential factors affecting house prices. Correlation analysis showed a strong positive relationship between these features and the target variable. Furnishing status also played a significant role, with unfurnished houses generally having lower prices compared to furnished properties.

Visualizations were created to better understand the data. A histogram of house prices highlighted the overall price distribution, while a correlation heatmap helped identify the strongest relationships between features and price. Feature importance visualizations further emphasized the significance of area, bathrooms, and other structural characteristics in determining property value.

Among the evaluated models, the machine learning approach demonstrated reasonable predictive capability. The Linear Regression model achieved an R^2 score of approximately **0.65**, indicating that it explained about **65% of the variation in house prices**. This suggests that the selected features provide meaningful information for predicting property values, although additional variables and more advanced feature engineering could further improve performance.

Overall, the project successfully demonstrated the complete machine learning workflow: data exploration, preprocessing, model training, evaluation, and interpretation. The results indicate that housing prices can be predicted with moderate accuracy, and that property size, bathroom count, and furnishing status are key drivers of market value.